VULNERBILITY ASSESSMENT OF GNSS L1 C/A SIGNAL USING SOFTWARE DEFINED SPOOFING ATTACK

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Dr. Salma Zainab Farooq

ABSTRACT

Global Navigation Satellite Systems (GNSS) structured interference, also known as GNSS Spoofing, is one of the main threats to satellite based navigation system that can result in the distraction, destruction, apprehension and even hijacking of the navigating body. Such type of threats needs to be studied thoroughly in order to develop appropriate countermeasures and defense mechanisms to make GNSS more reliable and secured. However, studying the effect of GNSS spoofing requires a spoofing scenario. In traditional methods, usually GNSS signal simulators are used for the construction of counterfeit data and generation of signal. But the cost of hardware equipment is usually extremely high and in addition to that, hardware simulators have fixed algorithm which implies that they cannot be modified if needed. For this research, a software based GNSS spoofer is developed in MATLAB environment and the signal generated by the spoofer is verified by GNSS receivers and smartphones. Furthermore, different types of errors such as ionosphere, troposphere, Doppler shift and code phases have been modeled to make the signal look exactly as if it is coming from actual satellites. Finally, the generated signal is stored in a binary file and transmitted using USRP N210. The signal is verified with a GNSS receiver first and then a smartphone later.

Keywords – GNSS Spoofing, GPS L1 C/A Code Simulation, Simplistic Spoofing Attack, GPS L1 C/A, USRP N210, GNSS Interference

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LIST OF ABBREVIATIONS

ABAS	Aircraft Based Augmentation System
ASIC	Application Specific Integrated Circuit
AGNSS	Assisted GNSS
AWACS	Airborne Early Warning and Control System
AOA	Angle of Arrival
ATC	Air Traffic Control
AGC	Automatic Gain Control
ADC	Analogue to Digital Converter
BPSK	Binary Phase Shift Keying
C/A	Course/Acquisition
CCDIS	Crystal Dynamics Data Information System
C/N0	Carrier to Noise Density Ratio
DLL	Delay Lock Loop
ECEF	Earth Centered Earth Fixed
ECI	Earth Centered Inertial
FIR	Finite Impulse Response
FLL	Frequency Locked Loop
GNSS	Global Navigaton Satellite System
GPS	Global Positioning System
GLONASS	Global Orbiting Navigation Satellite System
I	In-Phase Carrier Signal
IF	Intermediate Frequency
IGS	International GPS Service

L1	Link 1
L2	Link 2
LNA	Low Noise Amplifier
LO	Local Oscillator
NCO	Numerically controlled oscillator
PLL	Phase locked loop
PPS	Precise point positioning
PRN	Pseudorandom Noise
P(Y)	Precise
Q	Quadra-phase carrier signal
RMS	Root mean square
RF	Radio Frequency
SA	Selective availability
SNR	Signal to Noise Ratio
SPS	Standard Positioning Service
Std Dev	Standard Deviation
TEC	Total Electron Content
UTC	Coordinated Universal Time
WAAS	Wide Area Augmentation System
WGS-84	Word Geodetic System 1984

1. Introduction

Global Navigation Satellite System (GNSS) is one of the most widely used mean of positioning, navigation and timing (PNT) by various sectors including both civil and military. The system was launched initially for military applications only but later its services were made available to civilian users. Due it its maximum coverage, availability and passive service almost every sector is relying on GNSS for PNT services whether it be grid stations, time servers, banking and finance, or autonomous vehicles. Apart from that it provides continuous positioning without requiring any input from the end user which reduces the power requirement and cost of the user equipment. This is one of the main reasons why it is so popular mean of navigation. GNSS uses at least 4 satellites for the calculation of user position at any point on earth. Satellites orbiting in the outer space orbit send their own position, velocity and clock information to the user in the signal which is extracted by the receiver and used for the determination of range between the corresponding satellite and the user receiver. Range from multiple satellites ultimately results in user position coordinates. However, there are multiple error sources, such as ionospheric errors, tropospheric errors, multipath, and manmade interferences, which results in added biases and delays to the signal that dilutes the performance of the system. All these errors have been discussed in the next chapter in detail along with their contributing sources. Moreover, removal of these errors is extremely important in order to perform accurate navigation and get desirable outcomes. As mentioned, GNSS signals are highly vulnerable to errors from various natural and manmade sources. Even though most of these errors can be modeled and removed during the signal processing, manmade interference is something intentional and requires proper attention. There are two types of artificial interference i.e. jamming and spoofing. Jamming is the generation and transmission of signal with high power with an intention of completely disrupting the communication between the satellite and the receiver while spoofing is the transmission of counterfeit signal with forged information in order to deceive the victim device.

Figure 1.1 Global Navigation Satellite System

1.1 Problem Statement

While GNSS jamming poses a threat to the system as it blocks the communication between the satellites and the user equipment, it can easily be detected and can even be mitigated with some hardware and software level countermeasures. However, GNSS spoofing is a more serious issue as most of the time the receiver is not even aware of the presence of the attacking signal and can easily fall prey to such type of attacks. Due to this more and more research work is needed to be done in order to develop countermeasures for different types and levels of spoofing attacks.

However, for studying various anti-spoofing techniques, one needs to create an attacking scenario first which requires costly hardware equipment.

1.2 Motivation and Objective

The main motivation behind this research is to develop a software algorithm for GPS L1 C/A spoofing signal that is easy to implement in MATLAB environment and the generated signal can be transmitted with the help of any commercial of the shelf hardware equipment. The algorithm developed during this research provides a full control over the various parameters of the signal such as signal power, channels, time and delays etc. which can easily be varied according to the needs and applications of the researcher. This research will also enable students and scientists create a spoofing attack scenario within a controlled lab environment using low cost transmitting devices. Furthermore, it can also be used to study the behavior of the GNSS receivers during a spoofing attack. Apart from that, the generated signals will also be helpful for development and verification of software based GNSS receiver as the parameters can be manually set and known to the researchers.

1.3 Organization of Thesis

The first chapter of this thesis provides an introduction, problem statement, motivation and objective of this research work followed by chapter 2 which discusses a detailed review of the early and modern navigation methods, an overview of GNSS, GNSS segments, signal structure and a detailed literature review on GNSS threats i.e. jamming and spoofing and various types of spoofing attacks have been discussed in this chapter. Additionally, Chapter 3 includes the methodology and experimental setup and discuss the theory behind the generation of spoofing signal along with mathematical model. Chapter 4 includes results and discussion followed by chapter 5 which contains conclusion and future work.

2. Background and Literature Review

Accurate navigation, positioning and timing has always been the prime concern of mankind. In early days, localizing oneself was a cumbersome task especially in the sea due to the lack of reference points. On the other hand, land navigation was somehow easy due to the availability of reference points such as buildings, trees, squares, monuments, and flags etc. Accurate navigation in the sea was common problem for the sailors due to which most of the time ships were kept near the shoreline to track the previous and forthcoming position. In order to solve this problem people started working to find a solution to this common problem. As a result, many attempts were made to make the sea navigation not only possible but easier at the same time. During this time, people learned the importance of accurate time keeping in navigation so construction of an accurate time keeping device on a vessel became a new center of attention that could keep accurate time and remain undisturbed by the tides of the ocean during the voyage of the ship [1]. Furthermore, different means of navigation were invented including celestial navigation [2], dead reckoning [3], compass, etc. Although all these navigation methods were self-contained, but they were not reliable due to various reasons such as in bad weather condition, the celestial navigation system wouldn't simply work.

The quest for a reliable navigation system continued and with the passage of time and advancements in technology, satellites were launched that completely revolutionized navigation and positioning. The first satellite based navigation system was launched by the US Navy called "Transit" in September 1960 which helped the naval ships, and the Polaris ballistic missile submarines accurately perform navigation and positioning [4]. Besides that, it was also used for the hydrographic and geodetic surveying. Transit was replaced by the Navstar's Global Positioning System (GPS) which was the first Global Navigation Satellite System (GNSS). GPS was initially launched for military applications but later on its service was also made available for civilian applications which completely changed the position, navigation, and timing on earth. It consists of 32 satellites that are kept in the medium earth orbit. These satellites transmit passive signals towards the earth 24/7 which are received by receivers and decoded to extract navigation solution [5]. The best thing about GNSS is that it doesn't require any input from the user thus, the power requirement is very low. Due to the passive positioning and timing information, GNSS is widely used in various sectors for example GNSS based route planning, traffic and fleet management, navigation instruments for hiking and camping, drones and autonomous vehicles are only some of the land applications whereas it is quite effective for aerial and maritime applications as well.

2.1 Global Navigation Satellite Systems (GNSS)

GNSS is a broad term that is collectively used for four major global constellations including the US GPS, Russia's GLONASS, and Galileo of the Europe and BaiDou of the china. All of these systems are fully fledge functional and are providing global coverage to the users at any part of the word. Users on earth need at least 4 satellites for the calculation of position in which three satellites are used to extract the x, y, and z coordinates while the 4th satellite is used to correct the user receiver clock bias [6]. Each satellite transmits a navigation message that includes the orbit information of that particular satellite along with the clock information. This information is encoded in a specific pattern that is mentioned in the interface control document (ICD) [7] of each system. The receiver, upon receiving the signal from the satellite, demodulates the signal first and then decode the message data according to the ICD [8]. It then calculates the satellite position and range between the receiver and the satellite from which the signal is received. Once the range and the satellite positions are calculated, the output is used in positioning equations in order to extract the unknown values of the user location and time.

2.2 GNSS Segments

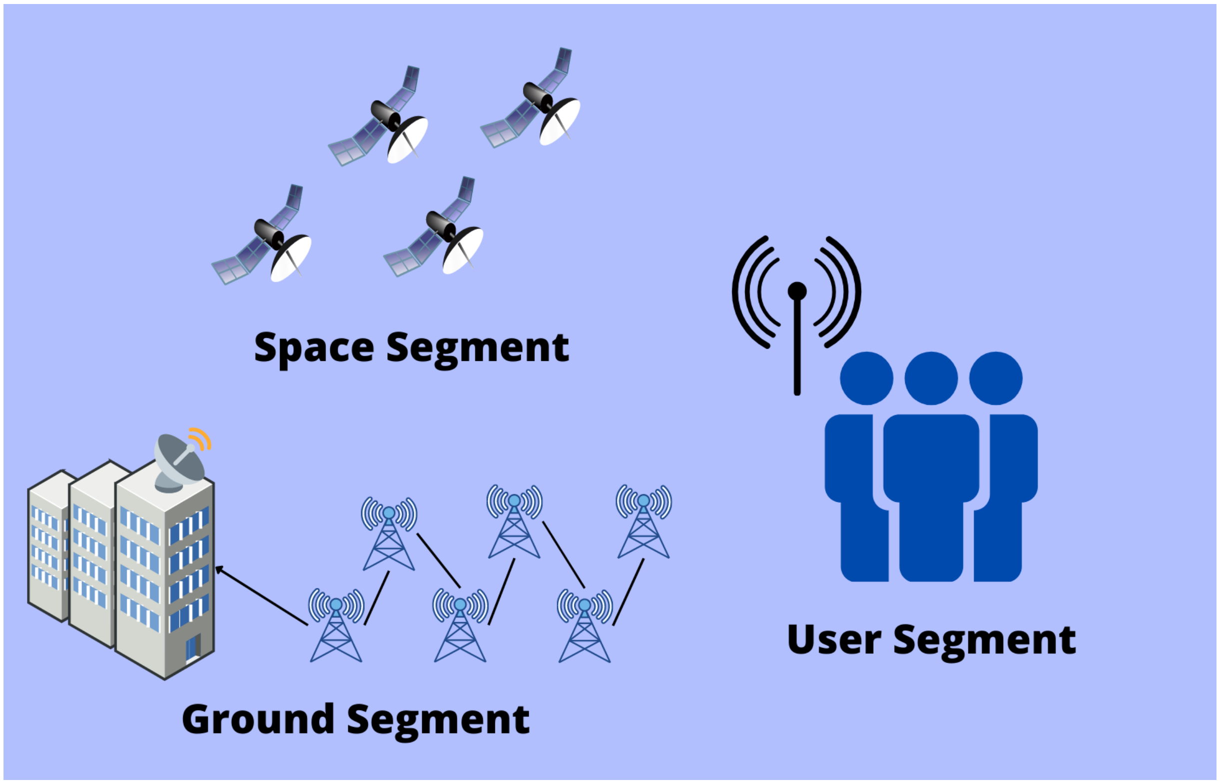

Figure 2.1 GNSS Segments

GNSS has three main segments, and each segment is responsible for their own functions. The layout of these segments is as follow [9]:

- Space Segment
- Ground Segment
- User Segment

2.2.1 Space Segment

The space segment of GNSS consists of the satellites and their payload. These satellites are usually kept in medium earth orbit with an altitude ranging from 20,000 km to 26,000 km. In case of GPS, there are a total of 32 satellites that are kept in 6 different orbits with an inclination angle of 55^{o}

[10]. These satellites are separated 60^{o} and the orbital period is one half sidereal day which makes 11 hours, 58 minutes, and 4 seconds. Due to this configuration around 8 to 12 satellites are visible to the user at any time on earth. Similarly, the GLONASS space segment contains 24 satellites that are kept in three orbital planes with eight satellites per orbit.GLONASS repeats its ground track after every eight days. In addition, Galileo uses 24/3/1 walker constellation [11] with 24 MEO satellites that are arranged in 3 orbital planes. The orbit inclination of Galileo is 56^{o} with respect to earth's equatorial plane. BeiDou features a unique space segment out of these four global constellations that has a total of 35 satellites, including 5 geostationary satellites, and 30 non-GSO. Furthermore, 27 of these non-GSO are in MEO while 3 satellites are kept in inclined Geosynchronous orbit (IGSO). The table 2.1 provides a brief overview of all GNSS constellations.

Table 2.1 GNSS Space Segment Specification

	GPS	**GLONASS**	**Galileo**	**BeiDou**
Total Number of Satellites	32	24	24	35
Orbit	MEO	MEO	MEO	MEO + IGSO
Orbital Inclination	55^{o}	64.8^{o}	56^{o}	55^{o} for both
Altitude	20,180 km	19,130 km	23,222 km	21,150 km
Period	11hrs, 58min, 4 sec	$8/17$ of sidereal day	14.08 hrs	7 days

2.2.2 Control Segment

The control segment of GNSS contains the command and control station, monitoring stations and ground antennas. The monitoring stations receives signals form the whole constellation using the ground antennas. It monitors the signals throughout the time and forward it to the command and control station. The command and control station estimates the clock for each satellite and orbital parameters which are also known as the ephemeris [12]. Command and control center is also equipped with an atomic clock that helps in the satellite clock correction. Apart from that, the responsibilities of a master control station also include housekeeping and anomaly resolution, navigation messages correction, satellite heath monitoring and signal integrity monitoring [13]–[16].

2.2.3 User Segment

The user segment consists of end user that receives the GNSS signal and use it for various applications. The best thing about GNSS is that it doesn't require extremely high power devices at the user end in order to receive the signal and perform navigation. Furthermore, users can use both hardware and software receivers for acquisition and tracking of the signal and navigation data extraction [17]. However, the accuracy of the GNSS can degrade highly depending upon the user location thus, it is backed by several augmentation systems in order to get the desired output and enhance its accuracy [18]. User equipment ranges from smartphones to high grade surveying and mapping receivers.

2.3 GNSS Signals and Signal Structure

GNSS uses radio waves in L-band for the transmission of its signals. The main reason behind choosing L-band for GNSS is that these signals can easily pass through clouds, rain, fog, storms, vegetation and can provide uninterrupted position and time information in just about any kind of weather [19]. In case of GPS, there are three main signal frequencies which are as follow:

- L1 = 1575.42 MHz
- L2 = 1227.45 MHz
- L5 = 1176.45 MHz

The signal structure consists of three main components i.e. carrier, code and data. GNSS uses CDMA multiplexing technique therefore, a pseudorandom noise code is assign to each satellite which combines the navigation data using direct spread spectrum sequence (DSSS) [17]. The combined C□D is then Binary Phase Shift Key (BPSK) modulated over the assigned carrier.

In case of GPS L1, the message structure is as follows:

- Carrier f_{L1} = 1575.42 MHz
- Code C/A = 1.023 MHz
- Data message = 50 bit/sec

Figure 2.2 provides an overview of the GNSS signal composition for L1 and L2 frequencies

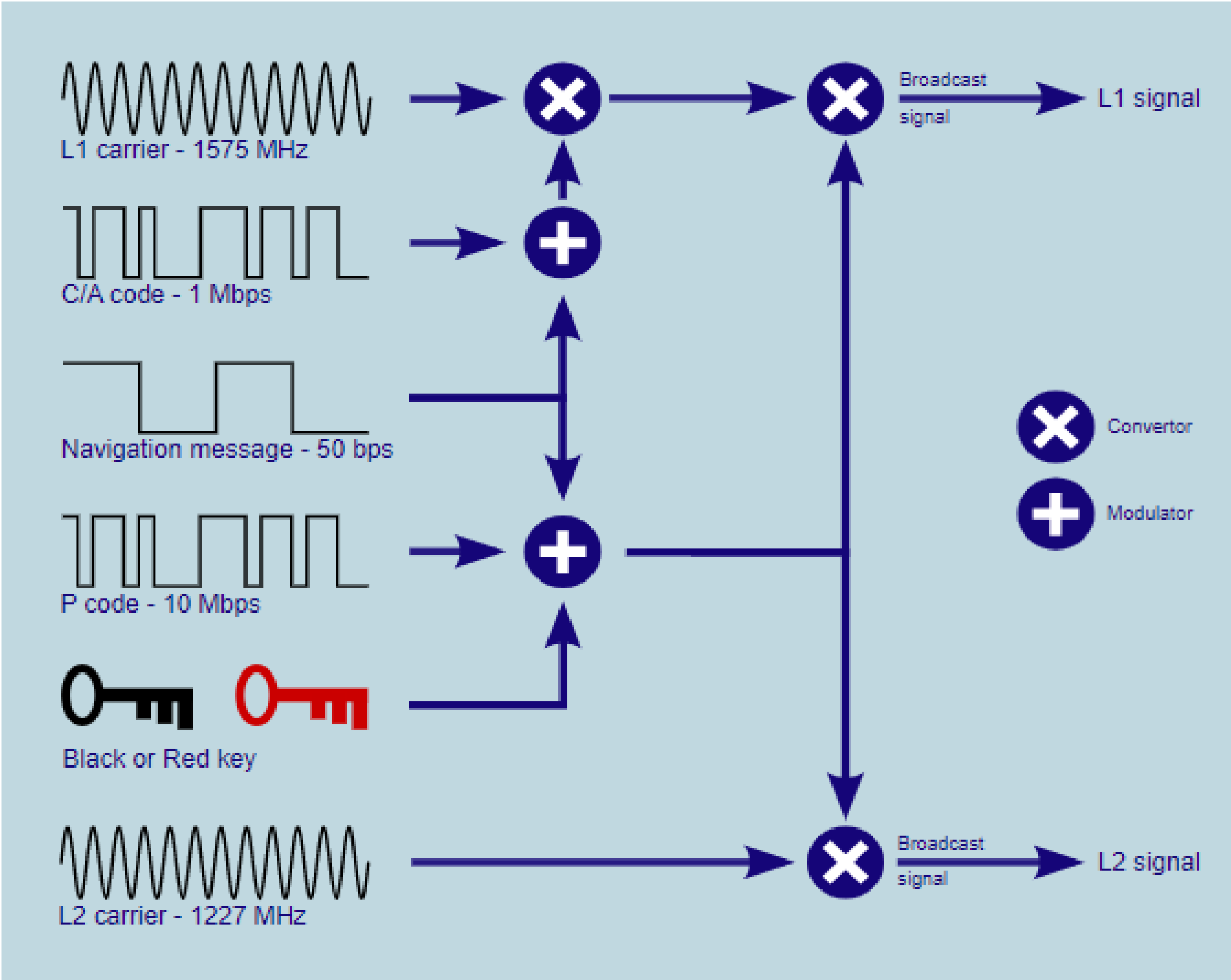

Figure 2.2 - GPS Signal Structure

As seen in the Figure 1 the code and data are modulo-2 added first and then modulated over the L1 carrier wave. Similarly, in case of the L2, the code and data are modulo-2 added and encrypted with a specific key that is only available to the authorized users [20]. The added code and message are then modulated over the L2 carrier.

2.4 Message Structure

The navigation message consists of specific components that includes the Keplerian elements to point satellite's position in space as well as some correction terms that are estimated by the control segment of the GNSS. The structure of GNSS message is usually explained in the ICD of the respective systems. In case of GPS, the legacy navigation message (LNAV) is broadcasted with a frequency of 50 bits/sec. The whole message is divided into frames or pages and each frame is divided into 5 sub frames. Each sub frame contains 10 words of data and each word consists of 30 bits. The starting of every sub frame word is telemetry (TLM) which contains a fixed preamble pattern i.e. 10001011 [21]. This preamble is used to mark the beginning of the sub frame and it is repeated after every 6 seconds. The second word of each sub frame is called hard over word (HOW). HOW contains a truncated version of time of the week called z-count which is the 17 most significant bits. The next two bits represents the information about anti-spoofing and selective availability. The next three bits represents the sub frame identity. In addition to the TLM and HOW there are 8 words in every sub frame that contains ephemeris data, clock data, satellite health and almanac [22]. A complete structure of GNSS Sub frames can be seen in the Figure 2.

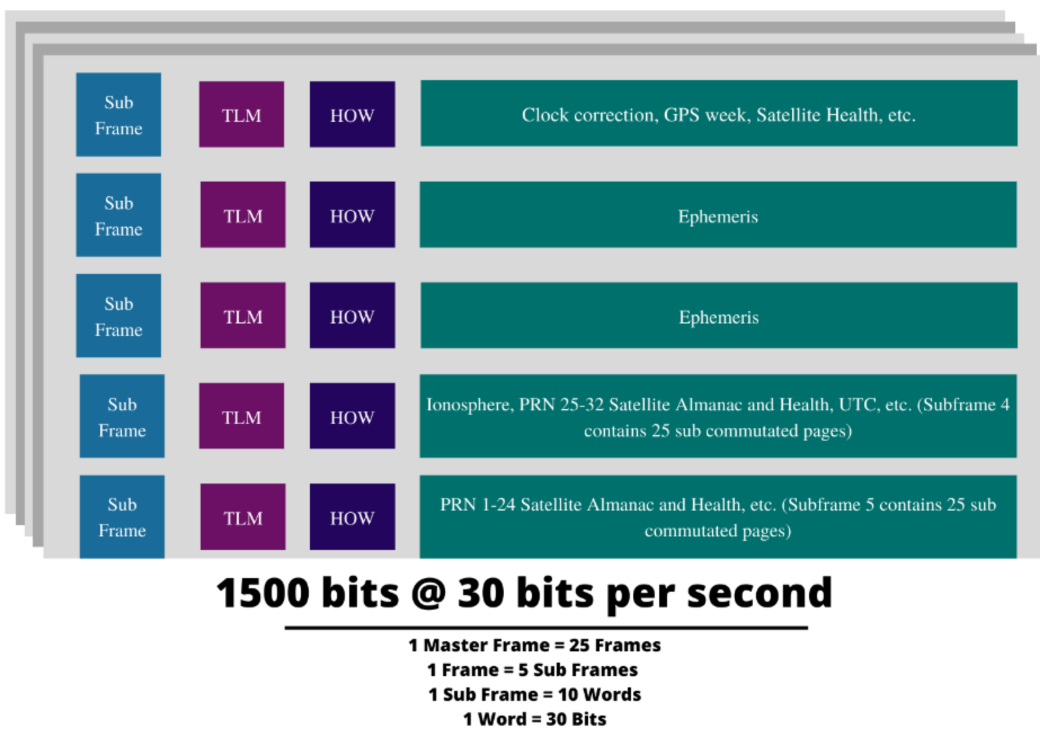

Figure 2.3 GNSS L1 C/A LNAV Message Structure

2.5 GNSS Errors

GNSS signals are prone to errors from multiple sources which degrades the overall performance of the system and dilutes the positioning accuracy. These errors need to be modeled and mitigated in order to achieve reliable positioning accuracy. GNSS errors can be divided into three main categories which are as follows:

2.5.1 Space Segment Errors

The space segment errors include the satellite clock error and ephemeris errors. Each satellite is equipped with a highly stable atomic clock that helps keeping an accurate time. However, drifts can occur even in the satellite clock and as a result biases are produced that needs to be fixed on time otherwise it can highly degrade the GNSS based positioning.

Besides that, satellite can deviate from its assigned path due to various influences from the outer space. The deviation of the satellite from its assigned orbit can result in a corrupted pseudorange [23]. The master control station continuously monitors the satellite clock and the ephemeris information obtained from the satellites. They correct the satellite clock and ephemeris as well as maneuver the satellite back to its assigned orbital path.

2.5.2 Channel Errors

After the transmission of the signals from each satellite it has to pass through a propagation channel in order to reach the receiving device. During the signal propagation it experiences delays from various sources which needs to be catered for in order to achieve the desired signal accuracy. There are various types of channel errors can degrade the overall signal accuracy and they are discussed below:

Ionosphere Errors: The ionosphere is a layer of the Earth's atmosphere that contains a lot of charged particles [24]. These particles can affect how GNSS signals travel from space to receivers on Earth. The ionosphere can cause two main problems for GNSS signals. The first is called ionospheric scintillation. This is when the charged particles in the ionosphere cause the signal to change speed as it travels through the atmosphere [25]. This can make the signal appear to move around erratically, which can make it hard for the receiver to track the signal. The second problem is called ionospheric delay. This is when the charged particles in the ionosphere slow down the signal as it travels through the atmosphere. This can cause the signal to arrive later than expected, which can throw off the timing of the receiver. Both of these problems can make it difficult for GNSS receivers to work correctly. However, there are some ways to mitigate these effects. One way is to use a dual frequency receiver which helps to calculate the total amount of delays occurring in the signal. Another way is to use ionospheric correction models that can help in the correction of the signal even with single frequency receivers [26].

Tropospheric Errors: Troposphere is the region below the ionosphere that also influence the signals up to a greater extent. Since the troposphere is not uniform and it varies from region to region therefore, these signals need to be modelled at regional level. There are two types of delays that the troposphere cause in the signal i.e. dry delays and wet delays. The dry component is more difficult to estimate as compared to the wet delays [27]. Fortunately, the dry component contributes the most delays in the signal. There are different models used for the mitigation of the tropospheric effects in the signal. Two of the most commonly used models in GNSS are the Saastamoinen model and the Hopfield model [28].

Sagnac Effect: Sagnac effect is a relativistic effect that occurs due to the rotation of the earth. It causes a delay proportional to the beam of the antenna. While this effect may not disturb the performance of the standard positioning however, it can degrade the performance of the precise positioning [23], [29]–[31]. A net rotation experienced by the user away from the satellite position will result in an increased propagation time. This needs to be corrected otherwise it will affect the positioning performance and can lead to a maximum error of 40m for GLONASS and GPS while in case of Galileo the error can reach up to 46 m.

Multipath Error: Multipath errors occur when the receiver receives a signal reflected from different surfaces such as buildings, vehicles, trees, etc. The multipath signals can cause interference and severely degrade the performance of the overall positioning of the GNSS [32], [33]. Multipath affects both the carrier and code based pseudorange measurements. Reflected signals can have significant variation in the magnitude depending upon the environment of the receiving device, signal processing at the receiver side, gain pattern of the antenna, and the satellite elevation angle.

2.5.3 Receiver Side Errors

Lastly, there are errors on the receiver side that contributes to the GNSS signal and degrades its performance. Receiver also adds an additional noise component to the signal during the conditioning and processing phase. This is mostly thermal noise which varies with the temperature of the components. Besides that, the receiver clock error is another factor that degrades the overall positioning accuracy. As we know the receiver clock is not as accurate as the satellite clock because it utilizes low quartz crystal clocks therefore [34], it will experience more drift as compared to the satellite. Fortunately, the receiver clock biases and drifts can be corrected with the help of GNSS clock which is why a fourth satellite is needed during the position calculation.

2.6 GNSS Interference

GNSS signals are extremely vulnerable to interferences from both manmade and natural sources. In case of natural sources, ionosphere and troposphere are the main sources of interference in the signal. Besides that, manmade interference sources include GNSS jamming and spoofing. Jamming is the transmission of the high powered signal towards a victim device with a goal of blocking the signal reception of the target device. It is mainly used for military purposes such as in war and combat situation [35]. Whereas spoofing is the transmission of the counterfeit signals towards a victim device with an objective to either distract, destruct, hijack or comprehend the target [36]. Both jamming and spoofing of the GNSS signals pose a serious threat to the system as they can descript the signals. For instance, a body that is fully relying on GNSS for its guidance and navigation can be distracted from its path. Among these two GNSS spoofing poses a serious threat because the victim device is usually not aware of the presence of the spoofing signals [37].

2.7 GNSS Jamming

GNSS jamming is the transmission of a powerful interference signal towards the victim device with an intention to jam the whole frequency spectrum of a specific band. It is mostly used in military scenarios such as jamming radio communication services, navigation signals etc. However, jamming cases in civilian sectors are also growing exponentially especially as the personal privacy devices are becoming more and more accessible to the common user. These personal privacy devices are small dongles that can even be powered with the help of a car's electronic system and can create a powerful interference signal that can disrupt the service from all major navigation satellite constellations. A well-known incident of GNSS jamming incident has been reported by the authors of [32] i.e. when the local area augmentation system's ground facility receivers were jammed by a mobile PPD passing nearby the New Jersey Airport [38].

However, this is not the only case as there are other cases that can be found in the literature. There was another famous incident when two US Navy ships were exercising near the San Diego Harbor and the technicians of the ships jammed radio signals [39]. The signals produced by these ships were so powerful that it disrupted the whole satellite based navigation system and several sectors were affected from this incident. During this period the local Air traffic control had to use backup system to control and maintain the flow of the air traffic.

The effect of GNSS jamming signal is based on two major parameters i.e. how powerful the transmitted signal is and how close the victim device is to the jammer. Furthermore, GNSS jamming can be characterized by two major characteristics which includes that jamming to signal ratio (J/S) and the central frequency of the attack signal. Since GNSS receivers uses carrier to noise ratio (C/N_0) thus the higher the C/N_0 the better will the signal quality and vice versa [39]. As the jammer comes near to the victim device the C/N_0 will decrease and the receiver will not be able to lock the incoming signals from the satellites [40]. It was confirmed experimentally by the author of [41] with the help of two drones. Furthermore, the authors also concluded that the civilian GNSS signals are more susceptible to Jamming as compared to the military signals. While Jamming is considered as one of the major threats to the GNSS infrastructure, it can be detected easily and can be avoided with the help of counter measures. However, GNSS spoofing is a much more serious threat as sometimes the victim device is not able to differentiate between the real and the counterfeit signals and it a successful spoofing attack can sometimes result in access to partial or even full control over the victim device [42].

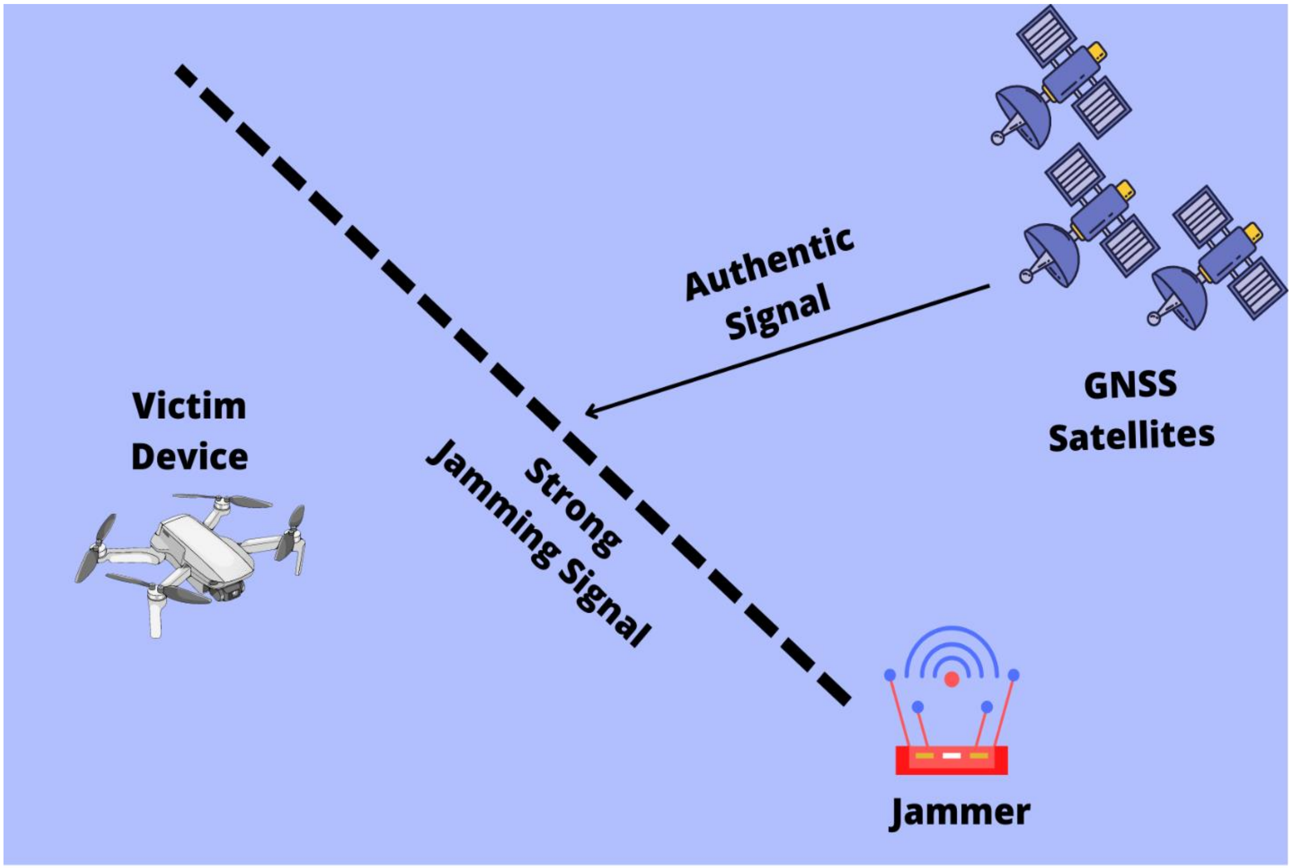

Figure 2.4 GNSS Jamming

2.8 GNSS Spoofing

GNSS Spoofing is the transmission of a counterfeit signal towards the victim device one of the following four objectives:

1. Distraction
2. Destruction
3. Hijacking
4. Apprehension

Civilian GNSS signals are open source and their details can easily be found in the interface control document of the respective system. These documents usually provide all the information about the signals such as their modulation schemes, their PRN generation sequence, their data structures and parameters included in the navigation message. This information can be reverse engineered to construct a GNSS-like signal and transmit it towards the victim device. The information in the message can be forged according to the will of the attacker. In contrast to civilian signals, the military GNSS signals are usually hard to spoof due to the lack of the availability of the information related to message and signal structure. Furthermore, these signals are encrypted in a sophisticated manner that helps them to be more secured as compared to their civilian counterparts [43].

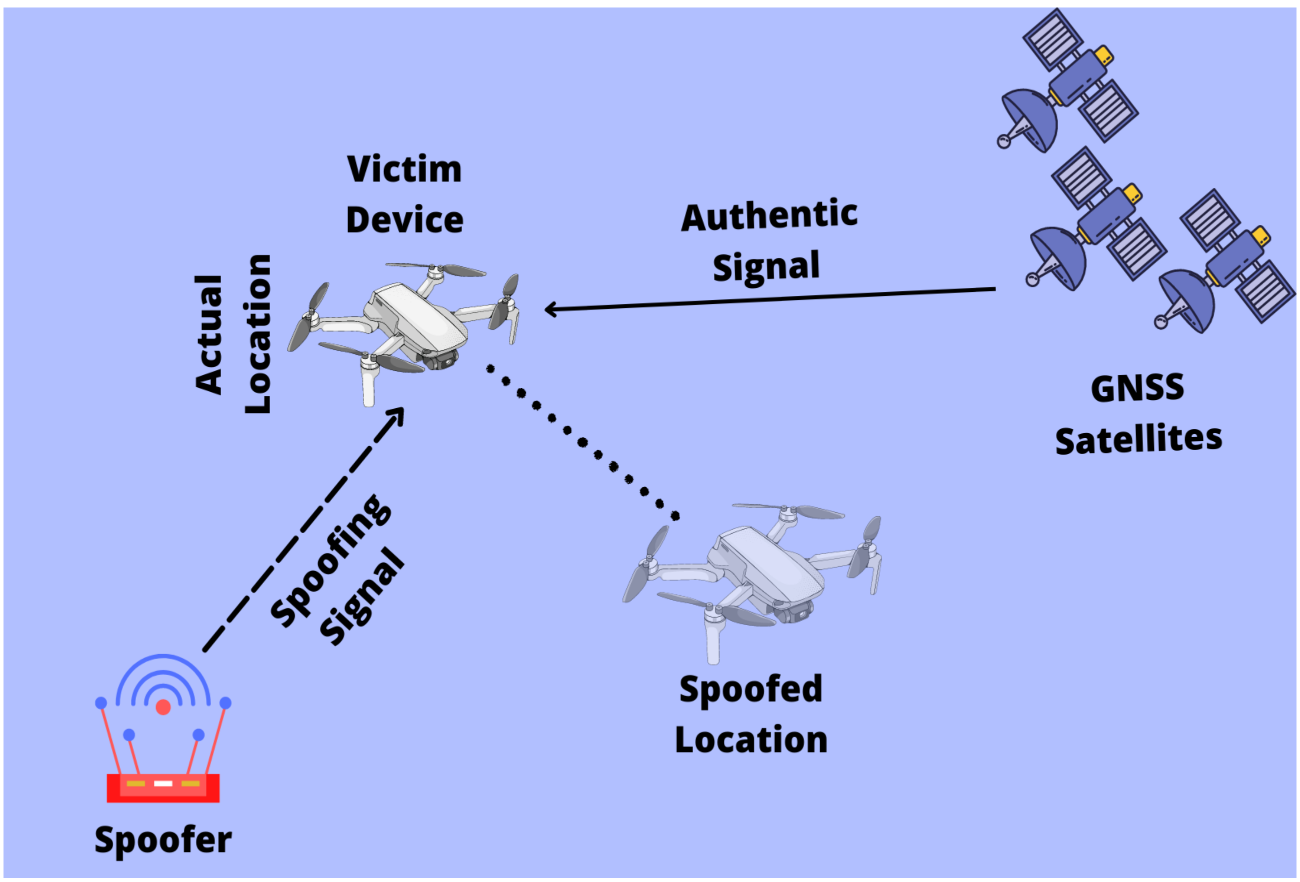

Figure 2.5 GNSS Spoofing

2.9 Types of GNSS Spoofing

Based on their characteristics, requirement of hardware and complexity levels, GNSS spoofing is divided into the following main types [44]:

1. Meaconing
2. Simplistic Spoofing
3. Intermediate Spoofing
4. Sophisticated Spoofing

Consider a receiver under attack, the signal can be modeled as under:

$$x_{RF}^{a}(t) = \sum_{l=0}^{N_s-1} S_{real,l}(t) + n(t) + a(t) \tag{2.1}$$

Here, N_s is number of visible satellites, $s_{real,l}$ represents the authentic signal component while the $a(t)$ represents the counterfeit spoofing signal and $n(t)$ is the total noise figure added to the signal. The signal transmitted from the spoofer can further be modeled as:

$$a(t) = \sum_{l=0}^{N_a-1} S_{spoof,l}(t) + n_{spoof}(t) \tag{2.2}$$

Here N_a is the number of satellites included in the signal, $S_{spoof,l}$ is the counterfeit signal component, and n_{spoof} is the added noise by the spoofer. A complete GNSS signal with proper amplitude and phase information can be written as:

$$S_{spoof,l}(t) = A_l(t)\hat{S}_{real,l}(t - \tau_l(t))\cos(2\pi\Delta f_l t + \Delta\theta_l) \tag{2.3}$$

In equation 2.3 A_l represents the amplitude of the signal, $\hat{s}_{real,l}$ is the estimate of the real signal with $\tau_l(t)$ a relative delay between the authentic and the spoofing signals. The amplitude and relative delays strictly depends upon the type of attack. Also, it should be noted that these values are not constant and varies from time to time. Figure 2.9 represents the hierarchy of spoofing scenario.

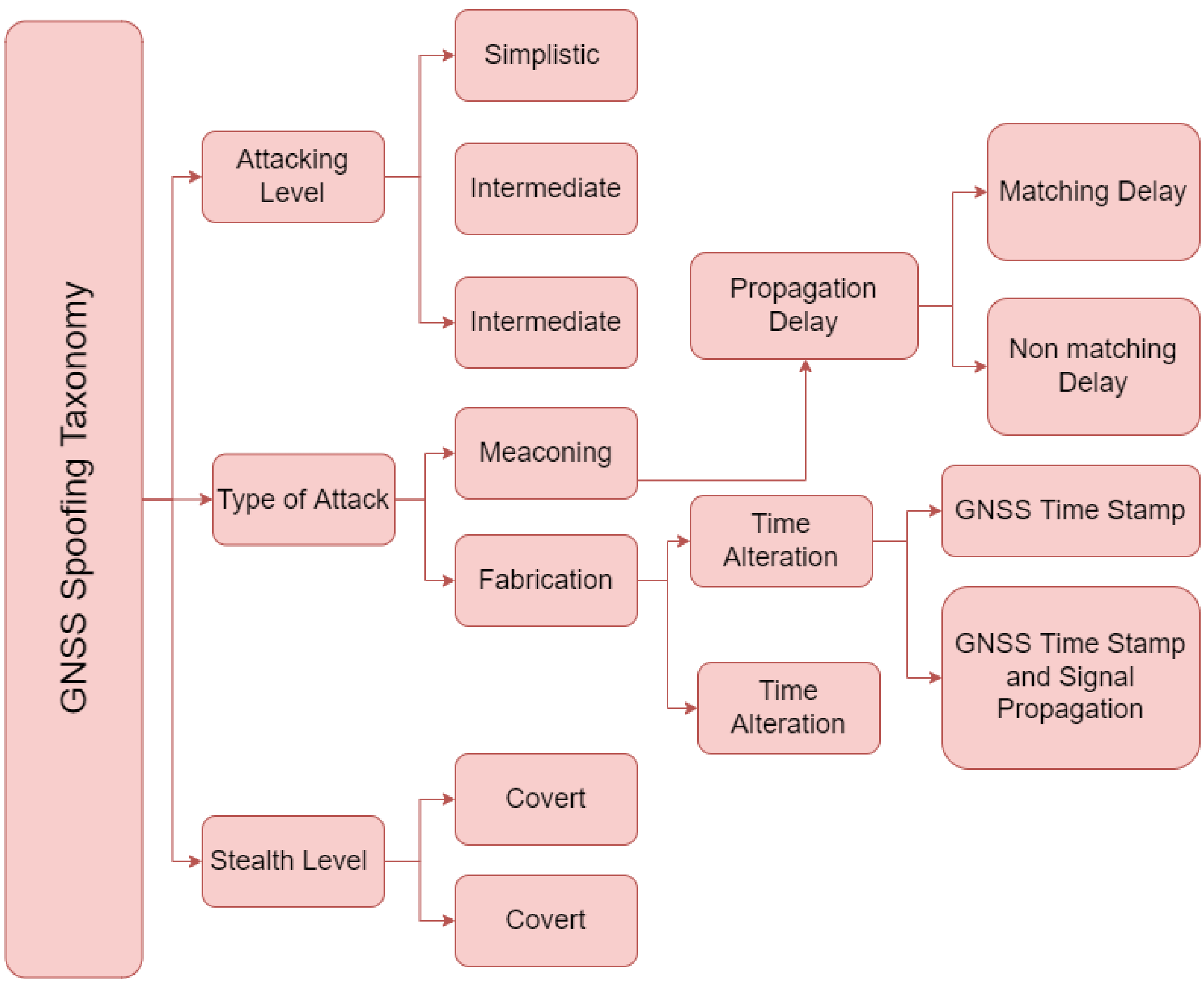

Figure 2.6 Hierarchy of GNSS Spoofing

2.9.1 Meaconing

Meaconing is a special type of GNSS spoofing attack which involves the recording of the real GNSS signal. This attack is carried out with the help of a signal repeater or a software defined transmitter. During meaconing a real GNSS signal is recorded for a specific duration at a certain location. This recorded signal is then appropriately sampled and retransmitted towards the victim device situated at some other location. While meaconing is one of the easiest type of spoofing attacks, it can easily be detected as the signals from the satellite as not aligned with the signals from the attacking module. Furthermore, the availability of the satellite at a specific position sometimes may not match the real satellite. In addition to this, during a meaconing attack, a relative delay is introduced to the signal that results in the signal arrives at the receiver with a positive delay. The main characteristic of the meaconing attack is the generation of the exact signals that are coming from the satellites and no residual modulation error or noise is introduced to the signal. However, the signal has a constant delay and an additional noise component is added by the attacking hardware that results in an increased noise level. This added noise figure is not negligible as well. The meaconing attack can be modelled mathematically as follow:

$$a(t)=\sum_{l=0}^{N_s-1} A_m s_{real,l}(t-\tau_m)+n_m(t) \tag{2.4}$$

Examples of GNSS meaconing are presented by [45], [46] where a detailed explanation is provided by the authors about the working principle, properties and drawbacks of meaconing attacks with the help of signal repeaters.

2.9.2 Simplistic Spoofing Attack

In case of simplistic spoofing attack, generally a GNSS signal simulator is involved that is used to generate a counterfeit signal and a transmitting device is connected in order to transmit the generated signals towards the victim receiver. Simplistic spoofing attack can easily be generated with both software or hardware signal simulators and the good thing about such type of attacks is that they attacker can adjust the signal according to the needs and requirements. However, the main issue with simplistic spoofing attack is that the signal produced by such techniques are not consistent and synchronized with the original signals coming directly from the GNSS satellites. Due to this, such types of attacks can easily be detected with the help of basic receiver level countermeasures. The mathematical model for the simplistic spoofing attack is given as follow:

$$S_{spoof,l}(t) = A_l(t)\hat{S}_{real,l}(t-\tau_l(t))\cos(2\pi\Delta f_l t + \Delta\theta_l) \tag{2.5}$$

It should be kept in mind that the replica signal generated through the simplistic spoofer, will not be consistent as well as they are affected by the discrepancies in the navigation message. Furthermore, the selection of satellites is merely a guess in such type of attacks so it can be different from the actual satellites visible to the victim receiver. Apart from that, signals generated through such method also contains a non-negligible residual modulation term that affects the performance of the overall attack. A suitable example for simplistic spoofing attack can be seen in [47] where the authors developed an algorithm for DJI Matrices 100 quadcopter. Another example of simplistic GNSS spoofing attack is can be studied in [48]. Furthermore, the authors of [48] also added that it is necessary to unlock the target receiver from the real signals first in order for the spoofing attack to work perfectly. Under such conditions, jamming the target device first can be helpful to disrupt the communication between the real signals from the satellite and the receiver.

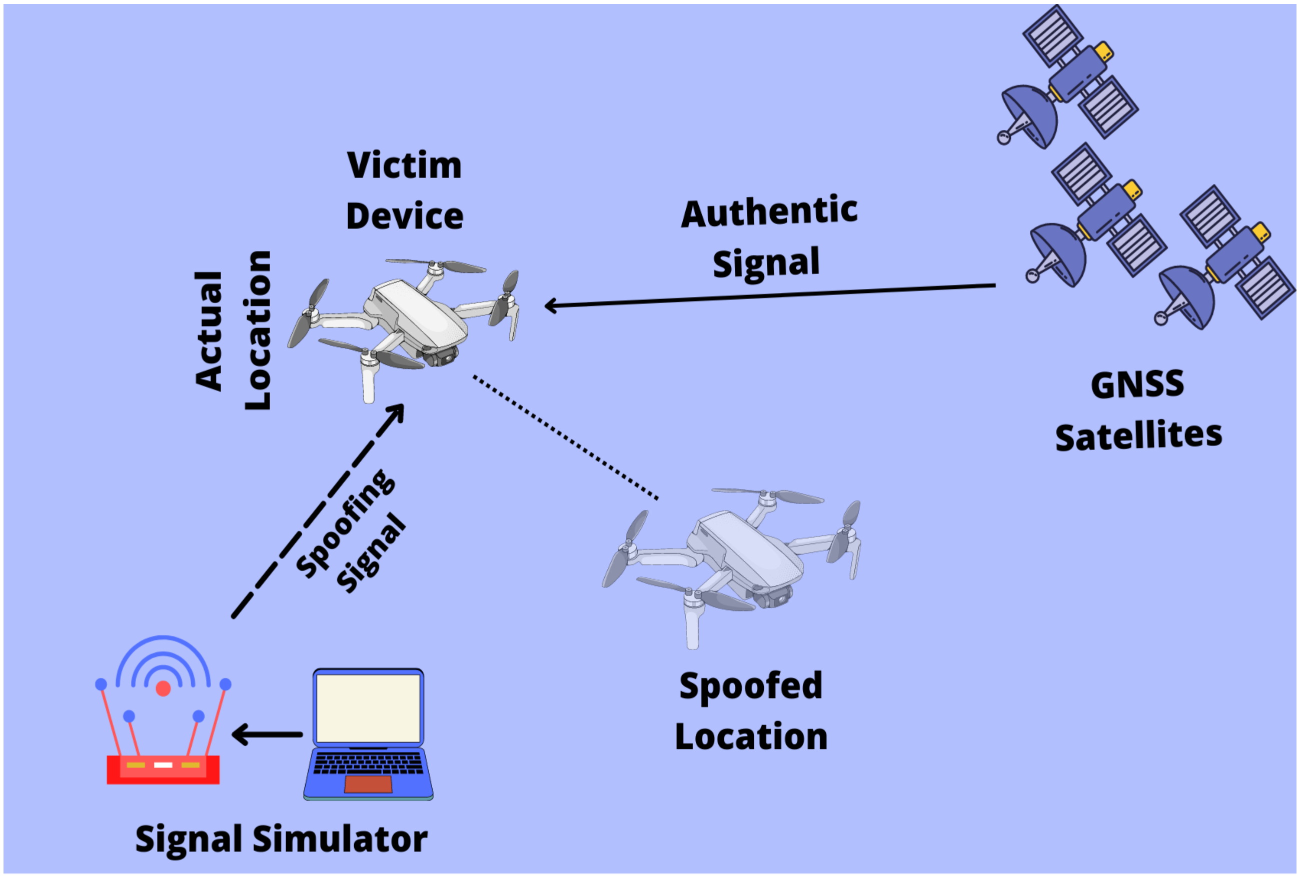

Figure 2.7 Simplistic GNSS Spoofing Attack Scenario

2.10 Intermediate Spoofing Attack

The intermediate attacking scenario is a slightly advanced version of GNSS spoofing and involves a receiver-transmitter architecture. The receiver module first locks the signals from the satellites directly and decodes it to obtain the time, satellite information such as ephemeris data, position, velocity, signal power etc. [44]. After that, it manipulates the signal with desired information, reconstruct and then transmit the signal towards the victim. Signals constructed through such types of attacks are usually highly synchronized as they are made on the basis of authentic signals [49]. In order to successfully carry out such an attack, the spoofing signals needs to be code-phase aligned and there should be at least a half chip synchronization with the real signals. This is the main characteristic of intermediate spoofing attack due to which it can't be easily noticed by the target receiver. Since the transmitted signals attacks each tracking channel of the victim receiver at the same time [50]. The signal model for the intermediate spoofing scenario is same as equation 2.5 however, the amplitude and the time delay are calculated relative to the real signals. Moreover, since the signals generated through this method are aligned therefore, there is no residual modulation term added during the attack. Such a spoofing attack can be studied in [49] where the author designed a software based intermediate spoofer and combined it with a transmitting front-end. In addition, the author of [44] modified a GNSS receiver for the generation of an intermediate spoofing attack signal. Moreover, several testbeds are used for the emulation of intermediate GNSS spoofing attacks as in [51]–[53]

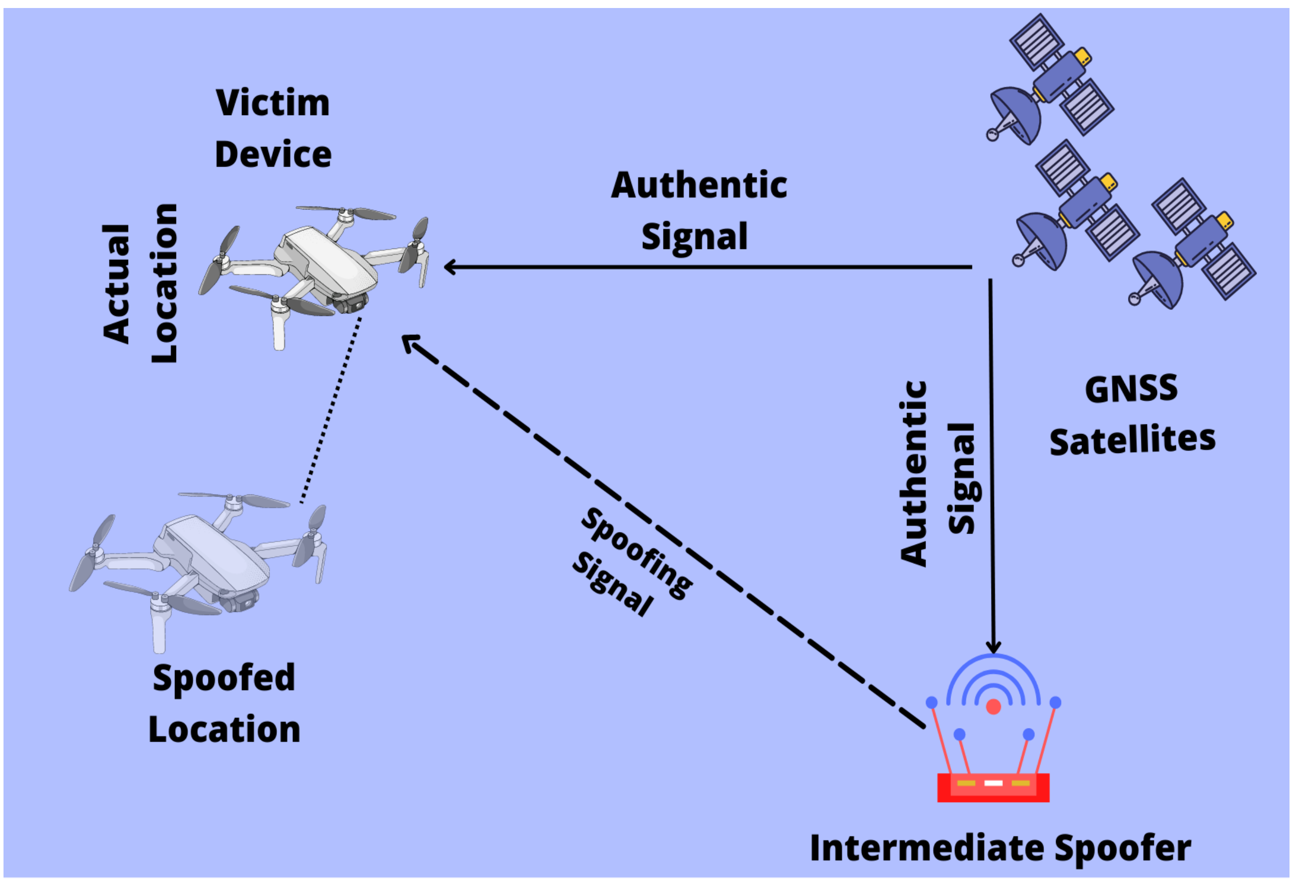

Figure 2.8 Intermediate GNSS Spoofing Attack Scenario

2.11 Sophisticated Spoofing Attack

In case of sophisticated spoofing attack scenario, the attacker usually uses coordinated receiver-spoofer architecture that are highly synchronized with each other. Such spoofing system is capable of generating a highly stable spoofing signal that is much harder to detect with basic and intermediate countermeasures [32]. Apart from that, the attacker has sub-centimeter level information about the target device's antenna phase center. In addition, such type of attacks can also successfully suppress the authentic signals in order to unlock the target receiver's tracking channels. The main characteristic of a sophisticated spoofing attack is the perfect synchronization with nearby attacking module of the chain. Signals generated by such an attacks have negligible frequency and phase difference. However, the construction of such type of spoofing mechanism is extremely difficult because of the accurate knowledge about the target receiver's position and velocity as well as synchronization of all spoofers in the chain. Due to these issues, the author of [54] has stated that no open literature has reported the construction and development of such type of receiver.

In literature there are several papers that presents different types of spoofing attacking scenarios. In [55] the author demonstrated spoofing attacks on a drone. Furthermore, in [56] an attack scenario was demonstrated on a yacht. Although GNSS spoofing is an illegal act, still more and more research work needs to be done in this area in order to identify the possible vulnerabilities in the signal and develop proper countermeasures to make the system more secured and reliable.

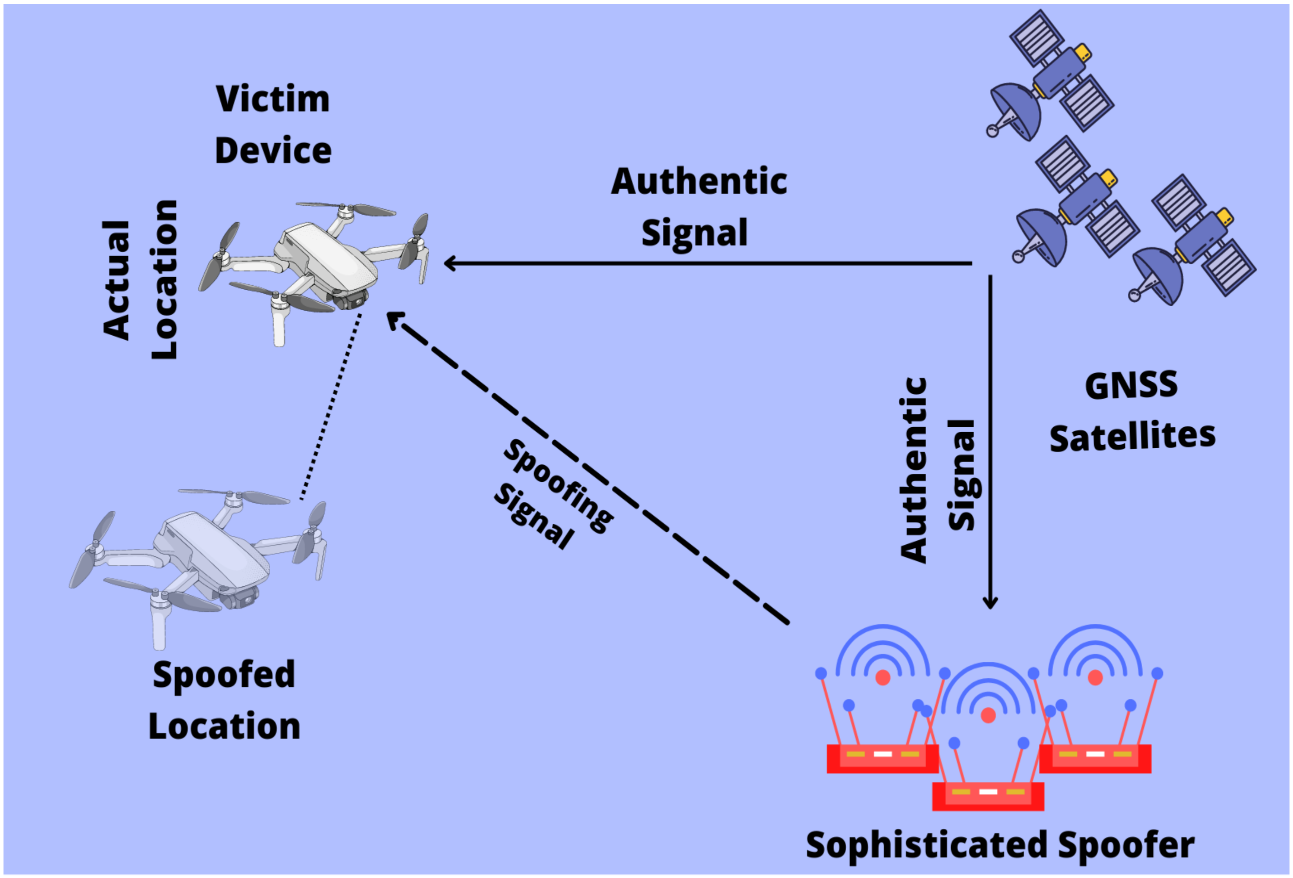

Figure 2.9 Sophisticated GNSS Spoofing Attack Scenario

Table 2.2 Comparison of Different Types of Spoofing Attacks Based on their level of detection and complexity

Type Of Attack	**Hardware Requirement**	**Context Requirement**	**Difficulty of Detection**
Meaconing	GNSS antenna, Low noise amplifier (LNA), RF Transmitter	Must be located in proximity to target receiver, LNA gain, and power level must be adapted according to the distance from the target.	Easy
Simplistic Spoofing Attack	Signal simulator, power amplifier, RF transmitter	The transmitting antenna must be close to the target	Very Easy
Intermediate Spoofing Attack	Modified receiver with transmitting front end	Target receiver antenna position and velocity must be accurately known	Hard
Sophisticated Spoofing Attack	Multiple phase-locked receivers-spoofers	Sub-centimeter level knowledge about target antenna phase center position and velocity	Very Hard

2.12 GNSS Spoofing Signal Construction Theory

Generation of Spoofing scenario requires either complex algorithms or expensive hardware equipment which makes study in this domain limited to specific research communities. However, by following a basic mathematical model GNSS spoofer can be derived to help in generation of the signal. The key elements to generate a counterfeit signal is the target spoofing location, satellite availability and position for the target spoofing location, user to satellite range, range rate, code and carrier phases and time. The satellite position can be guessed from the user position by reverse engineering the positioning algorithm and switching the known values of the user position with the unknown values of the satellite position. Furthermore, the ephemeris data can also be predicted however, it will be a very time-consuming task and will involve a lot of calculations. The best way is to record a GNSS signal for and extract the satellite ephemeris information for that recorded signal. Basically, the goal is to collect ephemeris data for the whole constellation. There are two ways to capture the whole constellation data i.e. by recording the data for 24 hours or by recording the data through multiple station and then merge it to a single file. In each of these situations, the ephemeris will be recorded for each satellite. It should be kept in mind that, ephemeris data from all 32 satellites is extremely important because the number of satellites varies from position to position. Therefore, if a target spoofing location is selected then visible satellites for that specific location needs to be selected carefully.

Once the data for the whole constellation is obtained, the next step is to extract the ephemeris information from the file and store in matrix for each satellite. This can easily be done in MATLAB or any other programming language. This data is used to calculate the satellite position. The steps for the calculation of the satellite position are given as under:

$$r = \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix} = \begin{bmatrix} a(\cos E - e) \\ a\sqrt{1-e^2}\sin E \\ 0 \end{bmatrix} \tag{2.6}$$

The above equation represents the satellite position vector with respect to earth. And by simplifying the above equation the following expression is obtained:

$$\| r \| = a(1 - e\cos E) \tag{2.7}$$

Here E varies with time while a and e are constant values while r is the geometric range between the satellite and the center of the earth. Let n represents the mean motion (i.e. the angular velocity of the satellite) and T represents the period of the orbit then we have:

$$n = \frac{T}{2\pi} = \sqrt{\frac{GM}{a^3}} \tag{2.8}$$

The combined value of GM is given in [5] to be 3.986005x10^14m3/s2. This value is used to compute the satellite position. Now let the elapsed time is given by t_0 so $\mu(t) = n(t - t_0)$. According to the kepler's equation the mean anomaly is related to the eccentric anomaly E as follow:

$$E = \mu + e\sin E \tag{2.9}$$

Now from equation (9) we get:

$$f = \arctan\frac{\eta}{\xi} = \arctan\frac{\sqrt{1-e^2}\sin E}{\cos E - e} \tag{2.10}$$

Equation **Error! Reference source not found.** connects the eccentric anomaly to the true and mean anomalies. Furthermore, the Cartesian coordinate of satellite is given as:

$$\begin{bmatrix} r_j^k \cos f_j^k \\ r_j^k \sin f_j^k \\ 0 \end{bmatrix} \tag{2.11}$$

The x, y, and z coordinates can be obtained by rotating the vector in the following sequence

$$R_3(-\Omega_j^k)R_1(-i_j^k)R_3(-\omega_j^k) \tag{2.12}$$

And finally the coordinates of the satellite at the time t_j can be derived as:

$$\begin{bmatrix} X^k(t_j) \\ Y^k(t_j) \\ Z^k(t_j) \end{bmatrix} = R_3(-\Omega_j^k)R_1(-i_j^k)R_3(-\omega_j^k) \begin{bmatrix} r_j^k \cos f_j^k \\ r_j^k \sin f_j^k \\ 0 \end{bmatrix} \tag{2.13}$$

Now at any given time of transmission the necessary values for the above equations can be obtained as mentioned in the [5] and using the corresponding values of each parameter from the ephemeris file results in the satellite position.

Using the above method, the satellite position is calculated for the whole constellation along with the velocity and accelerations. Once the satellite position is known for the target spoofing location, it can be used to calculate the azimuth and elevation angles for each satellite. These are extremely important values as they will help in determining which satellite is currently present at the target spoofing location. During this research, an elevation angle threshold of 10 degree was placed and all those satellites whose elevation angle was less than 10 degree were marked as not available. Once the available satellites are flagged, the next step is to allocate channels. Channel allocation is also very important as it allows the parallel processing of each satellite data and helps in the generation of the baseband signal. Channels are allocated to each PRN which also stores the corresponding data for each satellite i.e. the PRN, navigation message, C/A code, code and carrier phase, azimuth and elevation angles etc. After the allocation of channels, the range between the satellite and the target spoofing location needs to be determined by using the method mentioned in [Kaplan]. After the determination of the pseudorange, true range, and range rates, code and carrier phases needs to be calculated for each satellite. These two values are basically the core of the spoofing signal and by accurately modeling the Doppler frequency and code phases, the target victim device can easily be spoofed to the desired location, time and velocity. Once the code phases and the Doppler is calculated for each satellite, it is added to the signal in order to shift it and make it look like real. Furthermore, the ionospheric and tropospheric delays can also be added to further refine the signal along with the additive while Gaussian noise (AWGN). After perfectly modeling the signal for each satellite it is then sampled into I/Q file with a specific sampling frequency depending upon the transmitting device. During this research, the Ettus Research USRP N210 with a WBX daughterboard was used which supports a sampling frequency of 2.5 MHz and 16 bit baseband signal file. So the signal was modeled according to the specification of the transceiver.

3. Methodology and Experimental Setup

In order to carry out this experiment, Ublox ZED F9P receiver was used for the real signal data collection and later on for the verification of the spoofing signal. Ublox ZED F9P is a dual frequency GNSS receiver that supports signal from multiple constellations including SBAS satellites too. The receiver was installed in the astronomy observatory which is located at the top of Main block at the institute of space technology. The receiver was set to single positioning mode and was left for 24 hours for the data collection. However, it suffered from high interference, therefore, the location of the receiver was changed from Raza block to academic Block 7. It was then set to the same configuration and was left for 24 hours to collect the ephemeris data from all 32 GPS satellites. The data collection steps are as follow:

3.1 Installing U-Center and Ublox Drivers

Install U-Center software which is an open source software provided by the Ublox [57] and acts as an interface between the host computer and the connected receiver. This software provides a complete control over the receiver and the user can give several commands and configure different settings of the receiver directly from this software. Installation of U-center is quite straightforward and can be downloaded from the main Ublox website. Once the software is installed it will automatically install all the necessary drivers that are required to run the receivers from the Ublox. However, sometimes the host computer may not install the drivers automatically due to several issues. Under such conditions, the drivers can be downloaded from the official website [57] and installed manually using device manager on the host computer.

3.2 Establishing Connection and Configuring the Receiver

After successful installation of U-center, the next step is to establish connection between the receiver and host computer by selecting appropriate communication port on your computer. The baud rate should be set to auto in order to avoid any over or underflow of the data. After establishing connection, the receiver should be configured from the message view. During this research, the receiver was set to collect the RAWX and SFBX data only which contains the measurement data and the sub frame data respectively. However, the NMEA strings were turned off as they were causing issues due the file conversion. The receiver was set to single positioning mode and was left on recording mode for 24 hours.

3.3 UBX to RINEX Conversion

The output file obtained from Ublox ZED F9P was in UBX format and it needs to be converted into RINEX format [58] in order to read it in MATLAB and extract the data for each satellite. During file conversion, RTK Lib [59], an open source library, was used but no RINEX navigation file was obtained even after several tries and settings. Later on Emlid studio [60] was used for the conversion of the file and this time it successfully gave both RINEX nav and obs files. After this step, three different datasets were recorded for backup and to ensure that no satellite is missing from recorded data.

3.4 Reading RINEX File

RINEX file is ASCII format so MATLAB is capable of reading its content without any issue. A function “readrinex.m” was written for reading the ephemeris file, skipping unwanted information such as market name etc. and storing all the necessary data from the file into respective variables. Generally in post processing cases, mostly header of the RINEX file is skipped while reading its content. However, during this research, the Ion alpha and beta values along with the UTC data was also stored which were later used for adding ionospheric delays into the signal. Furthermore, readrinex function also extracted all the ephemeris parameters including the time at which the signal was recorded and saved it into eph structure. This information was then sorted for each satellite and was kept in an ascending order to make the satellite selection process easy during channel allocation process.

3.5 Select Ephemeris

After storing ephemeris information for navigation file, appropriate ephemeris needs to be selected for each satellite. Since the data was collected for 24 hours therefor various satellites ephemeris were repeating so this step is necessary in order to select the most recent ephemeris from file. In addition, the starting and ending time is observed from navigation file. Ephemeris information is selected in blocks with respect to time in such a way that there is one hour before and one hour after information with respect to the current time frame. In other words, if someone if the attack is being carried out at 12:30 am, the signal will have ephemeris data ranging from 11:30 pm to 1:30 am. Such a configuration helps resembling the ephemeris data update which occurs after every two hours. The time of ephemeris and time of clock data should be replaced with the most recent time values as this will make the signal look like real one. Apart from that, if the time is too old then there is a higher chance that some receivers will not lock the signal at all. Moreover, incorrect time will also result in incorrect satellite position and pseudorange information which can negatively affect the performance of the attack.

3.6 Calculate Satellite Position and Determine Visible Satellite

After updating toc and toe values in the ephemeris and selecting an appropriate block of data, satellite positions needs to be calculated for the whole constellation. Calculation of satellite position is already discussed in chapter 2 in detail so there is no need to discuss it again here. After calculation of satellite parameters such as position, velocity, acceleration, azimuth and elevation angles, visible satellite needs to be determined for the target spoofing location. For determination of visible satellites, a function was written in MATLAB named as satvisible.m. This function takes the satellite stats, azimuth and elevation angles and user position as its input where elevation of each satellite is checked and those whose elevation angle is greater than 5 degrees, is marked as visible.

3.7 Channel Allocation

During the next step, channels are allocated to each satellite which not only contains the PRN but all of its necessary data for processing as mentioned in the previous chapter. During this experiment, a total of 5 channels were created to reduce computational load during signal generation process. 5 to 6 channels are enough to successfully generated a signal and spoof the victim receiver. Pseudorange between each satellite in the channel and the target location was calculated and stored along with the geometric range, and pseudorange rate. Pseudoranges were then converted into code phases which will later be added to the signal as a delay to completely model it as if it is coming from the real satellite.

3.8 Generate Navigation Message

Generation of navigation message is one of the most important steps that needs to be taken carefully otherwise the target receiver will not lock the signals. The message structure of GPS L1

C/A is already given in the interface control document along with appropriate data decoding sequence. For this research, the message structure in the GPS ICD was reverse engineered in order to successfully encode data and stored into frames, sub frames, and words. Ephemeris data in RINEX format is in higher order and it needs to be scaled down first. The scale factors values are given in table xx. After scaling down the values it was converted into navigation bits and stored into words for each sub frame. There are a total of 5 sub frames in one complete frame and it takes 25 full frames to generate one master frame. Each sub frame is divided into 10 words and each word contains 30 bits so a total of 300 bits of data needs to be stored per sub frame. In addition, the first word of each sub frame is the TLM which stores a fixed 8-bit preamble (i.e. 10001011) that helps in the identification of the beginning of the sub frame at the receiver side. The second word of each sub frame contains a truncated version of z-count which is basically the time followed by bits for selective availability and anti-spoofing and sub frame ID which gives information about the sub frame number from which data is being extracted. Navigation data is stored in the first 3 sub frame while sub frame 4 and 5 contains almanac. Parity is calculated for each word in order to validate the navigation message. Parity calculation algorithm is mentioned in the figure 3.1.

Table 3.1 GNSS Message Parameters, Number of Bits and Scale Factors

Parameter	**Number of Bits**	**Scale Factor**
M_0	32	2^{-31}
Δn	16	2^{-43}
e	32	2^{-33}
$\sqrt{A}$	32	2^{-19}
Ω_0	32	2^{-31}

$\dot{i}_0$	32	2^{-31}
ω	32	2^{-31}
$\dot{\Omega}$	24	2^{-43}
IDOT	8	-
C_{uc}	16	2^{-29}
C_{us}	16	2^{-29}
C_{rc}	16	2^{-5}
C_{rs}	16	2^{-5}
C_{ic}	16	2^{-29}
C_{is}	16	2^{-29}
t_{oe}	16	2^{4}
IODE	8	2^{-43}

$$
\begin{aligned}
D_1 &= d_1 \oplus D_{30}^* \\
D_2 &= d_2 \oplus D_{30}^* \\
D_3 &= d_3 \oplus D_{30}^* \\
\vdots \quad & \quad \vdots \\
D_{24} &= d_{24} \oplus D_{30}^* \\
D_{25} &= D_{29}^* \oplus d_1 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6 \oplus d_{10} \oplus d_{11} \oplus d_{12} \oplus d_{13} \oplus d_{14} \oplus d_{17} \oplus d_{18} \oplus d_{20} \oplus d_{23} \\
D_{26} &= D_{30}^* \oplus d_2 \oplus d_3 \oplus d_4 \oplus d_6 \oplus d_7 \oplus d_{11} \oplus d_{12} \oplus d_{13} \oplus d_{14} \oplus d_{15} \oplus d_{18} \oplus d_{19} \oplus d_{21} \oplus d_{24} \\
D_{27} &= D_{29}^* \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_5 \oplus d_7 \oplus d_8 \oplus d_{12} \oplus d_{13} \oplus d_{14} \oplus d_{15} \oplus d_{16} \oplus d_{19} \oplus d_{20} \oplus d_{22} \\
D_{28} &= D_{30}^* \oplus d_2 \oplus d_4 \oplus d_5 \oplus d_6 \oplus d_8 \oplus d_9 \oplus d_{13} \oplus d_{14} \oplus d_{15} \oplus d_{16} \oplus d_{17} \oplus d_{20} \oplus d_{21} \oplus d_{23} \\
D_{29} &= D_{30}^* \oplus d_1 \oplus d_3 \oplus d_5 \oplus d_6 \oplus d_7 \oplus d_9 \oplus d_{10} \oplus d_{14} \oplus d_{15} \oplus d_{16} \oplus d_{17} \oplus d_{18} \oplus d_{21} \oplus d_{22} \oplus d_{24} \\
D_{30} &= D_{29}^* \oplus d_3 \oplus d_5 \oplus d_6 \oplus d_8 \oplus d_9 \oplus d_{10} \oplus d_{11} \oplus d_{13} \oplus d_{15} \oplus d_{19} \oplus d_{22} \oplus d_{23} \oplus d_{24}
\end{aligned}
$$

Figure 3.1 GPS Message Parity Calculation [5]

3.9 Generate Baseband Signal

After performing all of the calculation the last part is the signal generation. In case of GPS L1 C/A signals, the code and data are modulo-2 added which is also known as direct spread spectrum sequence. After the modulo-2 addition of the code and data, the resultant value is then BPSK modulated over the carrier. However, the L1 carrier is quite big for MATLAB processing environment therefore instead of a full carrier an intermediate frequency is chosen. The counterfeit signal was generated for a total duration of 300 seconds with a sampling frequency of 2.5 MHz however, sampling frequency is arbitrary and can be changed according to the transmitter specification. In our case, the sampling frequency of the transmitter was 2.5 MHz. Apart from that, the data resolution was kept as 16-bit which, again depends upon the requirements of the SDR for instance SDR from GreatScott Gadgets, HackRF One can support a file with 8-bit resolution. Similarly, the USRP can support files with 16-bit resolution. A baseband I/Q signal was generated with the above-mentioned specification and save to a file.

3.10 Signal Transmission

Once the signal is generated it is saved in a .bin file which is an extension for binary file format. This .bin file can later on be transmitted with the help of any SDR based transmitting device. USRP N210 with a WBX daughterboard was used as a transmitter. To avoid wasting time on driver installation and interfacing between USRP and windows platform, Ubuntu platform was used instead. It allowed direct communication between SDR and the operating system by quickly downloading all the necessary drivers by using terminal commands. Once the USRP was successfully configured, baseband I/Q file was transmitted with 1575.41 MHz frequency and USRP's stock clock was used as a reference oscillator. It was found that the stock oscillator of USRP is not very stable and can drift with time.

4. Results and Discussion

For this research, the spoofing signal was transmitted in a controlled environment in GNSS Lab at IST to avoid interfering with the Space and Upper Atmosphere Research Commission (SUPARCO) equipment which is near to IST. Furthermore, the signal was transmitted with different gains ranging from 0 to 75 dB and it was found that the most suitable gain of USRP for GPS L1 C/A signals is 75 dB as signals transmitted with such a gain were found to be consistent. However, at first the receiver was acquiring all the satellites in the signal but none of them were tracked. After further investigation, it was found that the Doppler was not perfectly modeled. After successfully modeling the Doppler frequency shift, this time the receiver was able to not only acquire the signal but also lock the respective satellites in the signal and get position from the extracted data.

4.1 Results with GNSS Receiver

4.1.1 Scenario Test 1: China

During the first test, the spoofing signal was generated for China with coordinates as under:

- **Latitude**: 30.28650348907680
- **Longitude**: 120.03266761583689
- **Altitude**: 95.91846689861268

The signal was tested and verified with GNSS-SDR [61] an open source software defined GNSS receiver library which allows the processing of baseband signals within Linux/Ubuntu environment. During this attack, the signal was generated for a total duration of 30 seconds and it was noticed that the receiver was unable to extract all sub frames for the signal and as a result a loss of lock occurred in multiple tracking channels. The signal duration was then increased to 36 seconds but still no message was received by the receiver. After generating the signal for a duration of 300 seconds, finally the GNSS-SDR was able to lock each satellite in the tracking channel and decode the navigation data from the baseband signal file. The decoded data was then saved into RINEX 2.11 file format which was processed in RTK Lib and KML File was generated and plotted on google earth for the verification of the results.

As seen in the figure 4.1 the receiver was kept in the academic block 7, GNSS lab at Institute of space technology.

Figure 4.1 True Location of the Receiver

When the signal was transmitted towards the receiver, the location switched from IST block 7 to China at exact same coordinates for which the signal was forged. Figure 4.2 represents the spoofed location of the receiver.

Figure 4.2 Spoofed Location during the Attack

During this scenario, the receiver was able to track a total of 5 satellites consistently as clearly seen in the figure below.

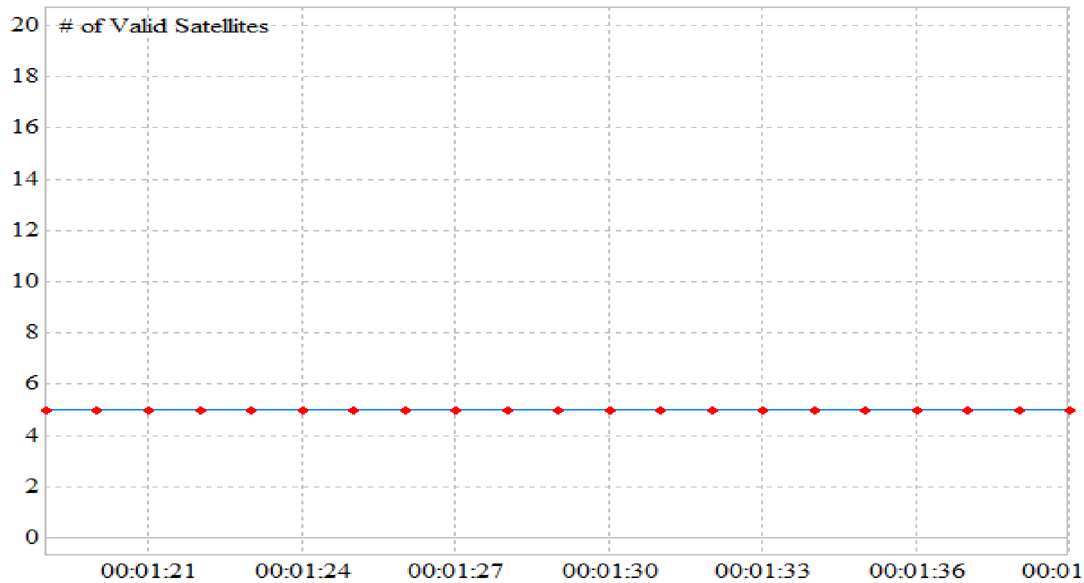

Figure 4.3 Number of Visible Satellites

Apart from that, figure 4.5 represents the ground track for the scenario. As seen in the picture, the ground track represents as if the receiver is in motion however, in reality it is a static position but since GNSS provides positioning at every epoch therefore, it can't pin point the exact same location at each epoch and hence variations in coordinates can be observed in the figure 4.5. This variation in coordinates is complete normal and can be expected with any commercial off the shelf GNSS receiver.

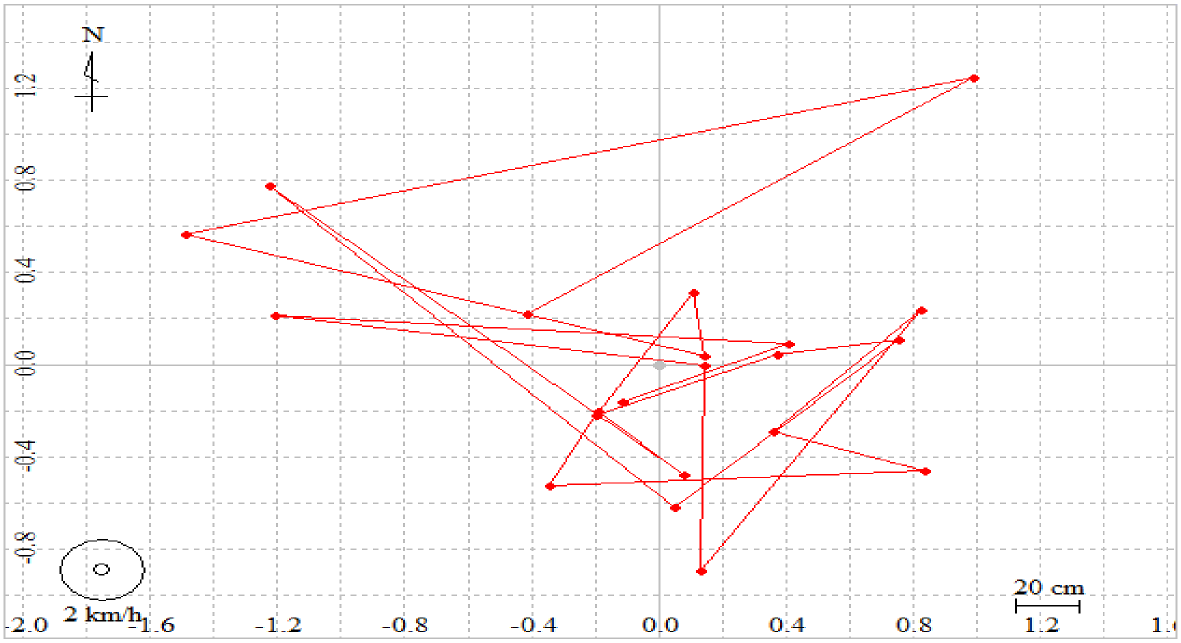

Figure 4.4 Ground Track of the Receiver

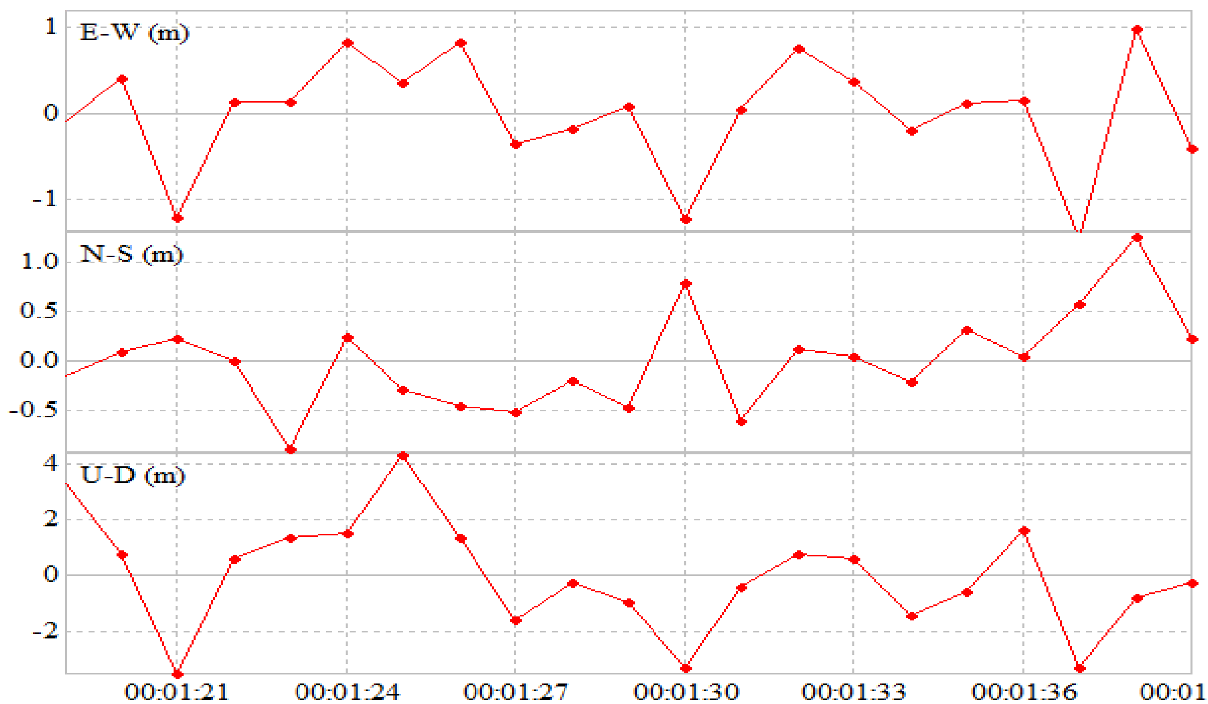

Figure 4.5 Variation in Coordinates Relative to Epochs

Furthermore, to verify the working of the algorithm another signal was generated for France and the signal was sampled and written to an I/Q binary file as mentioned before and it was tested with a GNSS receiver.

4.1.2 Scenario Test 2: France

During this test, the target spoofing location was one of the world's famous monumental point i.e. the Eiffel Tower which is located in France. The coordinates for the desired target location are given as follow:

- **Latitude:** 48.85842352427114
- **Longitude:** 2.29380265975847
- **Altitude:** 284.97946438752115

The result obtained for the location can be seen in figure 4.6, while the ground track, number of visible satellites for the receiver, and variation in coordinates can be seen in figure 4.7, 4.8, 4.9 respectively.

Figure 4.6 Spoofed Location of Receiver Showing Position as France

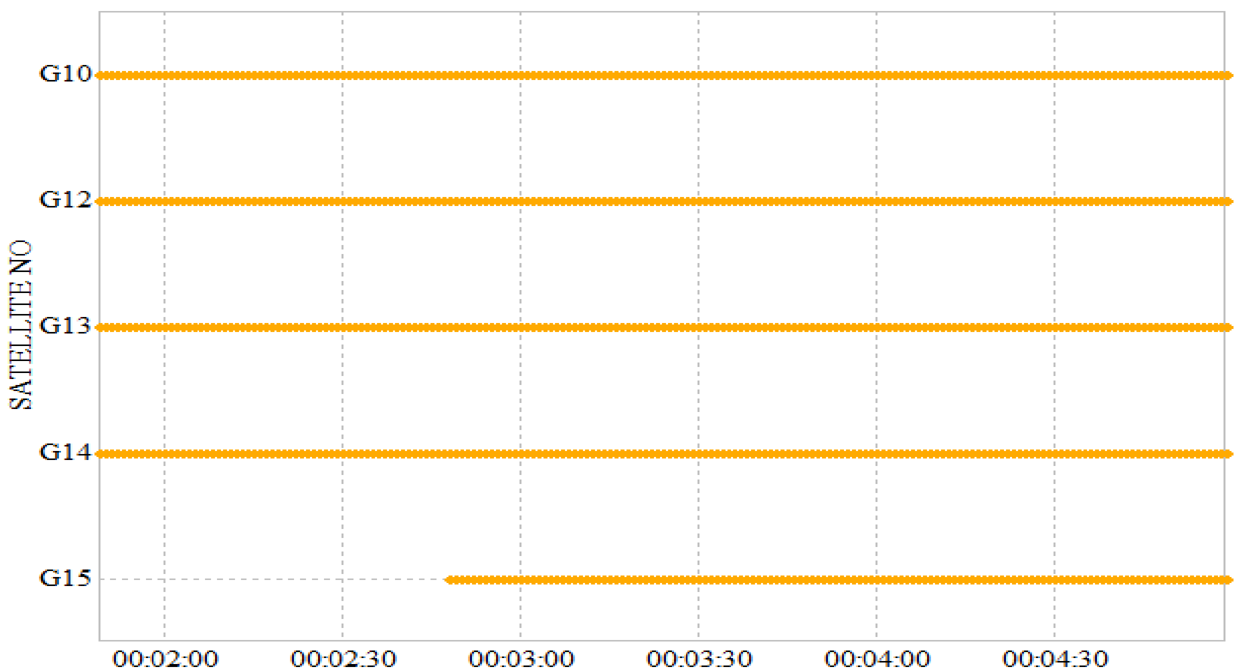

Figure 4.7 Number of Visible Satellites Tracked By the Receiver during the Attack

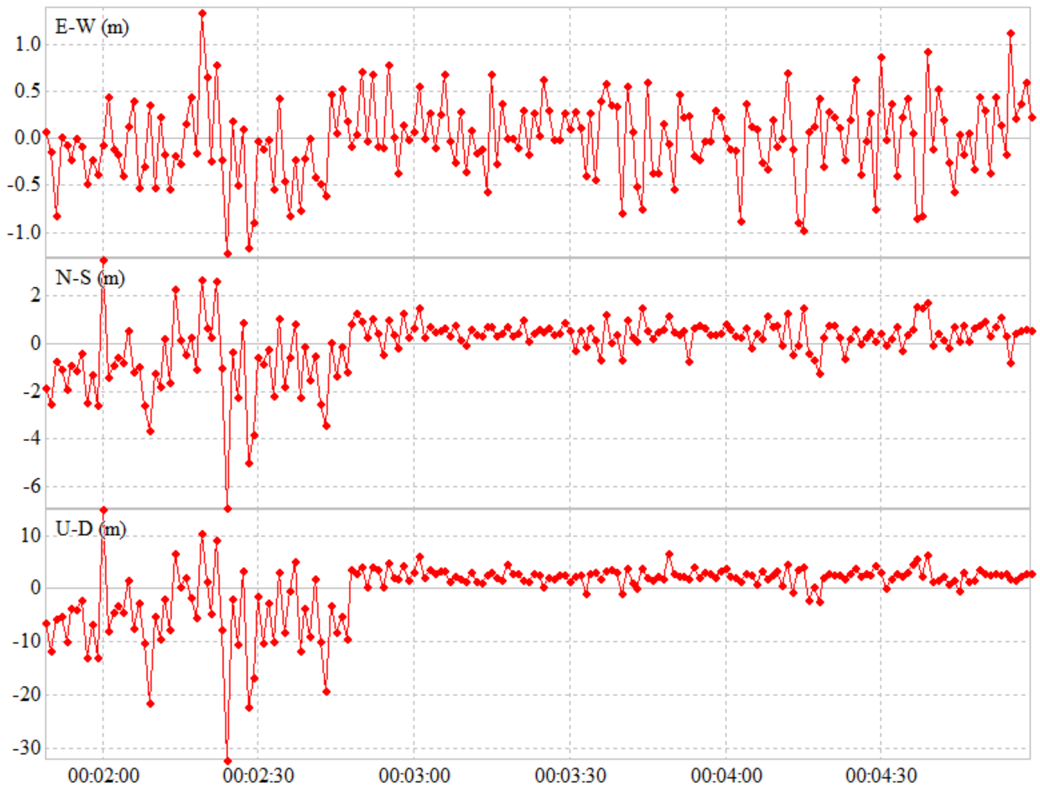

Figure 4.8 Variation in Coordinates Relative to Epochs

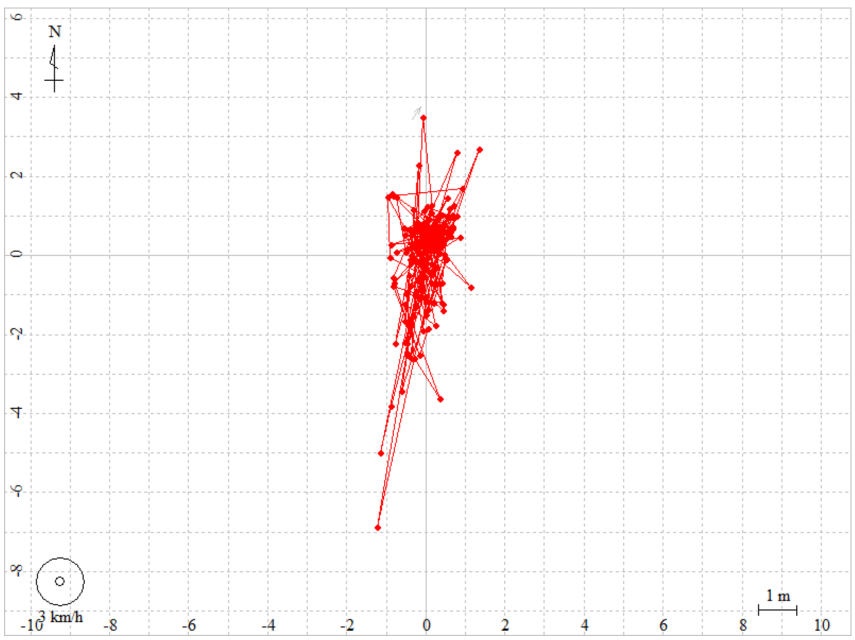

Figure 4.9 Ground Track of the Receiver during the Attack

4.2 Results with Smartphone

Apart from the GNSS receiver, the signal was also received on various smartphones to confirm the working of the attack. Three different smartphones were used for testing purposes i.e. the Realme 5 pro, Infinix Hot 11s and the Google Pixel 4 XL. Before the attacking scenario we were fully expecting the Realme 5 pro and the Infinix Hots to be spoofed by the signal while the Google pixel will show resistance to the attack. However, to our surprise the experimental results were quite different from what we expected. During a series of tests, we noticed that the Realme 5 pro was completely resilient to the attacks despite of turning off WiFi, mobile data and additional positioning settings such as Bluetooth and WiFi scanning while the Infinix Hot 11s was proved as the most vulnerable phone as it was the first one to lock the counterfeit signal and calculate a 3D fix followed by the Google Pixel 4XL. The results can be seen in the following figures.

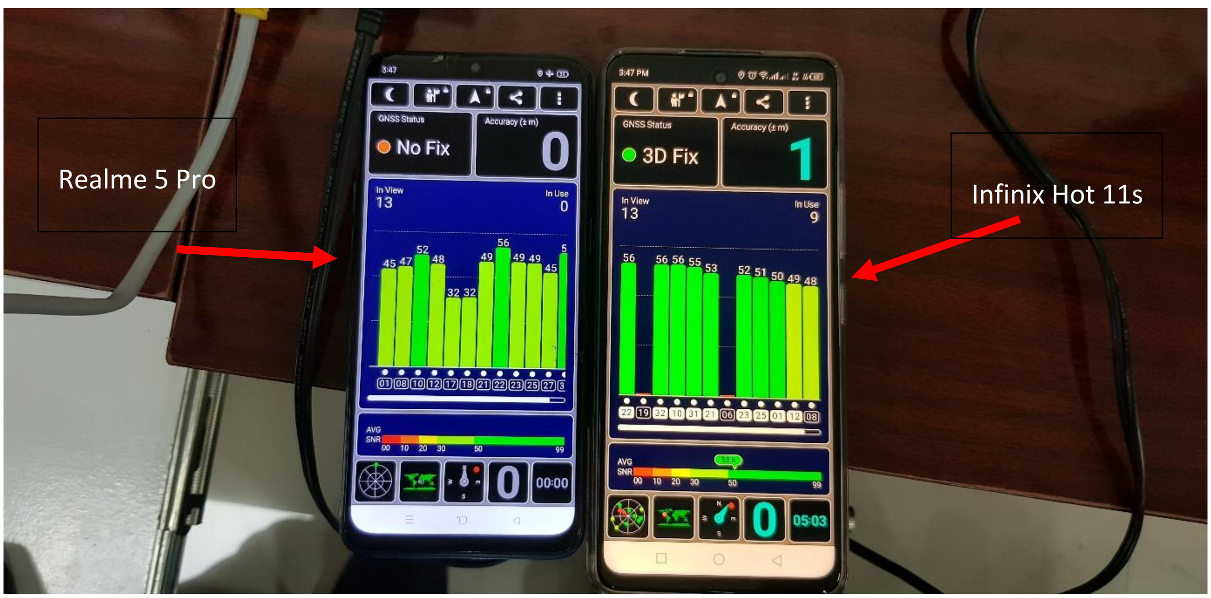

Figure 4.10 Realme 5 Pro (on Left) and Infinix Hot 11s (on Right) During the Attack

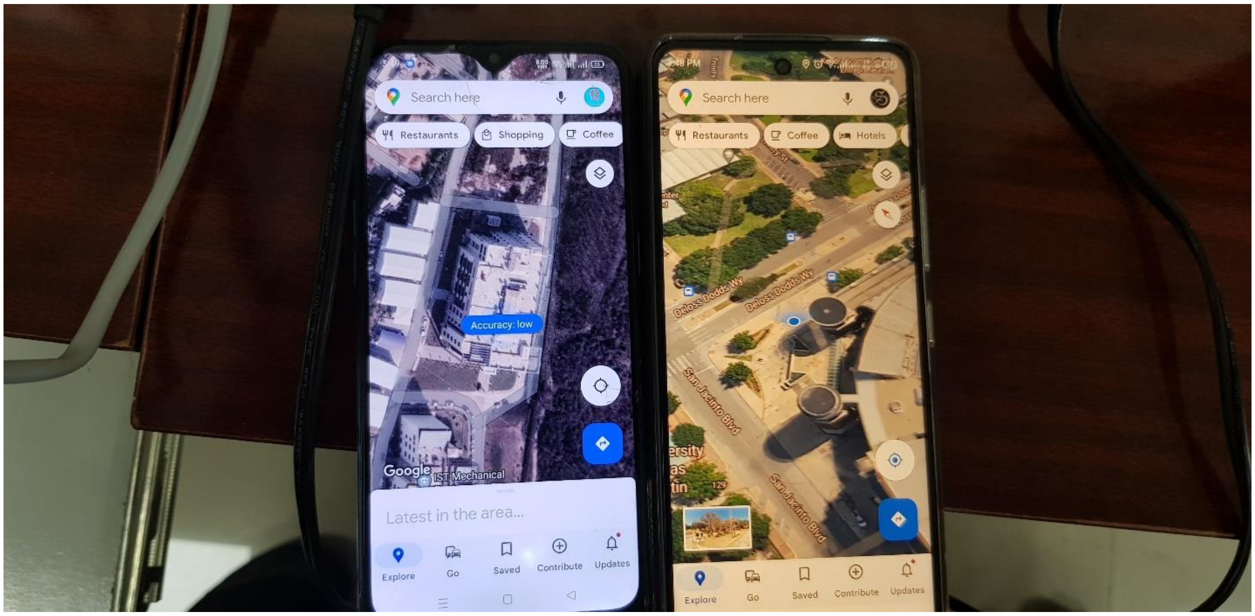

Figure 4.11 Realme 5 Pro showing Actual Position of the Receiver, Infinix Showing Spoofed Location

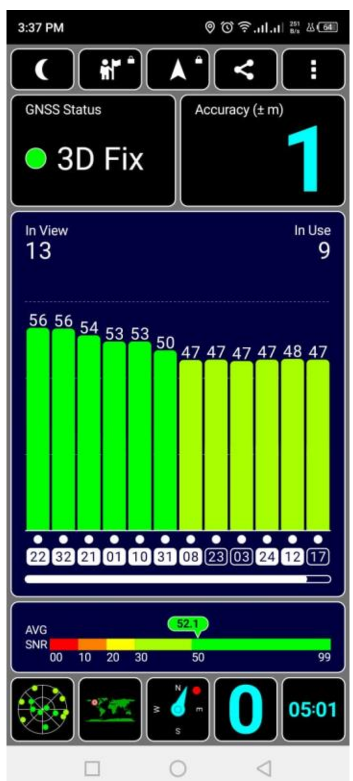

Figure 4.12 Screenshot of Google Pixel 4XL during the Attack

5. Conclusion

Global Navigation Satellite System is one of the best means of positioning, navigation and timing. But security isn't a built-in feature any of the GNSS whether it be GPS, GLONASS, Galileo or BeiDou. Apart from that, it is highly vulnerable to both manmade and natural interferences. During this research, GNSS L1 C/A vulnerability was studied with the help of structured interference also known as spoofing. A software based spoofing algorithm was designed for the generation of the signal which takes recent ephemeris values recorded from the real satellites and then forge a signal with desired time and coordinates. The generated signal was tested and verified on Ublox Zed F9P and three different Android based smartphones. Results can be seen in chapter 4 where the receivers showed multiple spoofing locations. However, it was also found that some smartphones are now working on GNSS security as there are multiple positioning modes in almost all new smartphones such as Wi-Fi and Bluetooth scanning and built-in IMU sensors of the smartphone. Furthermore, the security patches that are released for smartphones also seems to be improving with time in terms of integrity and positioning accuracy of the GNSS services that the phone is getting.

Spoofing attacks can be quite destructive in case of both military and civilian scenarios as more and more sectors are completely relying on GNSS for their positioning timing and navigation applications. Thus more and more research work needs to be done in both spoofing and anti-spoofing domains to timely discover the vulnerabilities in the signal and counter them successfully.

In conclusion, generating a spoofing scenario is not a difficult task provided that the attacker is aware of the signal and message structure. Furthermore, GNSS is strictly a time of arrival based system therefore, time plays a crucial role not only in the positioning but also in the generation of a counterfeit signal. It is because incorrect time will lead to corruption of several parameters such as week number, second of the week, pseudoranges, carrier and code phases and hence the signal can't be modeled accurately.

5.1 Future Research Directions

As mentioned, research in the spoofing domain is necessary in order to discover all of the vulnerability of the GNSS system and timely develop appropriate countermeasures. The following are some of the future research direction/ideas:

- In future, this research can be extended to multi-constellations and multi-frequencies as modern GNSS receivers are configured for multiple frequencies and constellations.
- Spoofing attacks can also be used for to identify the vulnerabilities RTK and DGNSS based receivers, time servers and integrated navigation system.
- Since GNSS is used to correct the errors that grows unboundedly in the INS therefore, I suspect it can also be turned as a parasite sensor and negatively configure the states of INS which can result in incorrect positioning during sensor fusion based navigation
- GNSS a spoofing scenario can be constructed based on the manipulation of the ephemeris data directly from the received signal. Under such conditions the attacker won't have to rely on the ephemeris data recorded from the real satellites.

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